

Airline Data Management and Analysis Using Power BI

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- Course : *Data Science (May 15th)*

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Problem Statement

The airline industry operates with numerous complexities, requiring effective data management and insights into flight schedules, passenger details, and ticketing systems. This project aims to analyze airline operations for improving efficiency and customer satisfaction.

Objectives

- To clean and prepare airline data.
- To build a relational data model using FlightID.
- To create calculated columns and DAX measures for insights.
- To design interactive dashboards with slicers, quick views, and drillthroughs.
- To implement Row-Level Security (RLS).

Dataset Description

Explain briefly:

- **Flight_Information** (FlightID, FlightNumber, Airline, Destination, Status).
- **Passenger_Information** (PassengerID, FlightID, SeatNumber).
- **Ticket_Information** (TicketID, FlightID, BookingStatus).

Tasks Completed (with Screenshots)

Task 1: Data Preparation & Cleaning

- *Removed duplicates.*
- *Handled missing values.*
- *Formatted columns.*

The screenshot displays the Power Query Editor interface. The main area shows a table with the following columns: FlightID, FlightNumber, Airline, Destination, and Status. The table contains 26 rows of data. The 'Applied Steps' pane on the right shows the 'Removed Columns' step, which has been applied to the 'flight_information' query. The step description indicates that columns 6 through 14 were removed from the original data.

FlightID	FlightNumber	Airline	Destination	Status
1	1001 FL1102	Airline D	Houston	On Time
2	1002 FL1435	Airline B	Chicago	On Time
3	1003 FL1860	Airline A	New York	Cancelled
4	1004 FL1270	Airline C	Chicago	Delayed
5	1005 FL1106	Airline C	New York	Delayed
6	1006 FL1071	Airline A	Phoenix	On Time
7	1007 FL1700	Airline C	Los Angeles	Cancelled
8	1008 FL1020	Airline C	Los Angeles	Delayed
9	1009 FL1614	Airline A	Los Angeles	Cancelled
10	1010 FL1121	Airline D	Chicago	Cancelled
11	1011 FL1466	Airline A	Phoenix	On Time
12	1012 FL1214	Airline D	New York	Delayed
13	1013 FL1330	Airline C	Houston	On Time
14	1014 FL1458	Airline C	New York	Delayed
15	1015 FL1087	Airline C	Houston	Delayed
16	1016 FL1372	Airline B	New York	Delayed
17	1017 FL1099	Airline D	Phoenix	Delayed
18	1018 FL1871	Airline B	Houston	Delayed
19	1019 FL1663	Airline B	Chicago	Cancelled
20	1020 FL1130	Airline A	New York	On Time
21	1021 FL1661	Airline B	New York	Cancelled
22	1022 FL1308	Airline A	Houston	Delayed
23	1023 FL1789	Airline A	Chicago	On Time
24	1024 FL1343	Airline B	Chicago	Delayed
25	1025 FL1491	Airline D	Phoenix	On Time
26	1026 FL1413	Airline D	Chicago	Cancelled

Untitled - Power Query Editor

Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Replace Values Data Type: Whole Number Merge Queries Append Queries Text Analytics Vision Combine Files Azure Machine Learning

Queries [2]

flight_information

ticket_information

Table.Distinct(#"Removed Columns", {"TicketID"})

TicketID	FlightID	BookingStatus
1	5001	1178 Pending
2	5002	1078 Confirmed
3	5003	1117 Cancelled
4	5004	1120 Cancelled
5	5005	1137 Cancelled
6	5006	1162 Pending
7	5007	1076 Pending
8	5008	1035 Cancelled
9	5009	1001 Cancelled
10	5010	1040 Cancelled
11	5011	1064 Pending
12	5012	1150 Cancelled
13	5013	1060 Cancelled
14	5014	1064 Confirmed
15	5015	1093 Confirmed
16	5016	1072 Pending
17	5017	1011 Cancelled
18	5018	1105 Cancelled
19	5019	1014 Confirmed
20	5020	1060 Pending
21	5021	1030 Confirmed
22	5022	1035 Confirmed
23	5023	1165 Confirmed
24	5024	1005 Confirmed
25	5025	1083 Cancelled
26	5026	1123 Cancelled

3 COLUMNS, 50 ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name

ticket_information

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Removed Columns

Removed Duplicates

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Untitled - Power Query Editor

Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Replace Values Data Type: Text Merge Queries Append Queries Text Analytics Vision Combine Files Azure Machine Learning

Queries [3]

flight_information

ticket_information

passenger_information

Table.RemoveColumns(#"Removed Duplicates", {"Column4", "Column5", "Column6", "Column7", "Column8", "Column9", "Column10", "Column11",

PassengerID	FlightID	SeatNumber
1	1	1161 38A
2	2	1157 24D
3	3	1141 30B
4	4	1046 17E
5	5	1035 29D
6	6	1134 10A
7	7	1082 10A
8	8	1115 20E
9	9	1197 34E
10	10	1047 2E
11	11	1153 43C
12	12	1194 48C
13	13	1010 47A
14	14	1056 23C
15	15	1030 16D
16	16	1109 40D
17	17	1005 25C
18	18	1119 32C
19	19	1033 27E
20	20	1118 32B
21	21	1065 19E
22	22	1146 5B
23	23	1177 28B
24	24	1011 22E
25	25	1085 6A
26	26	1026 5A

3 COLUMNS, 100 ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name

passenger_information

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Filtered Rows

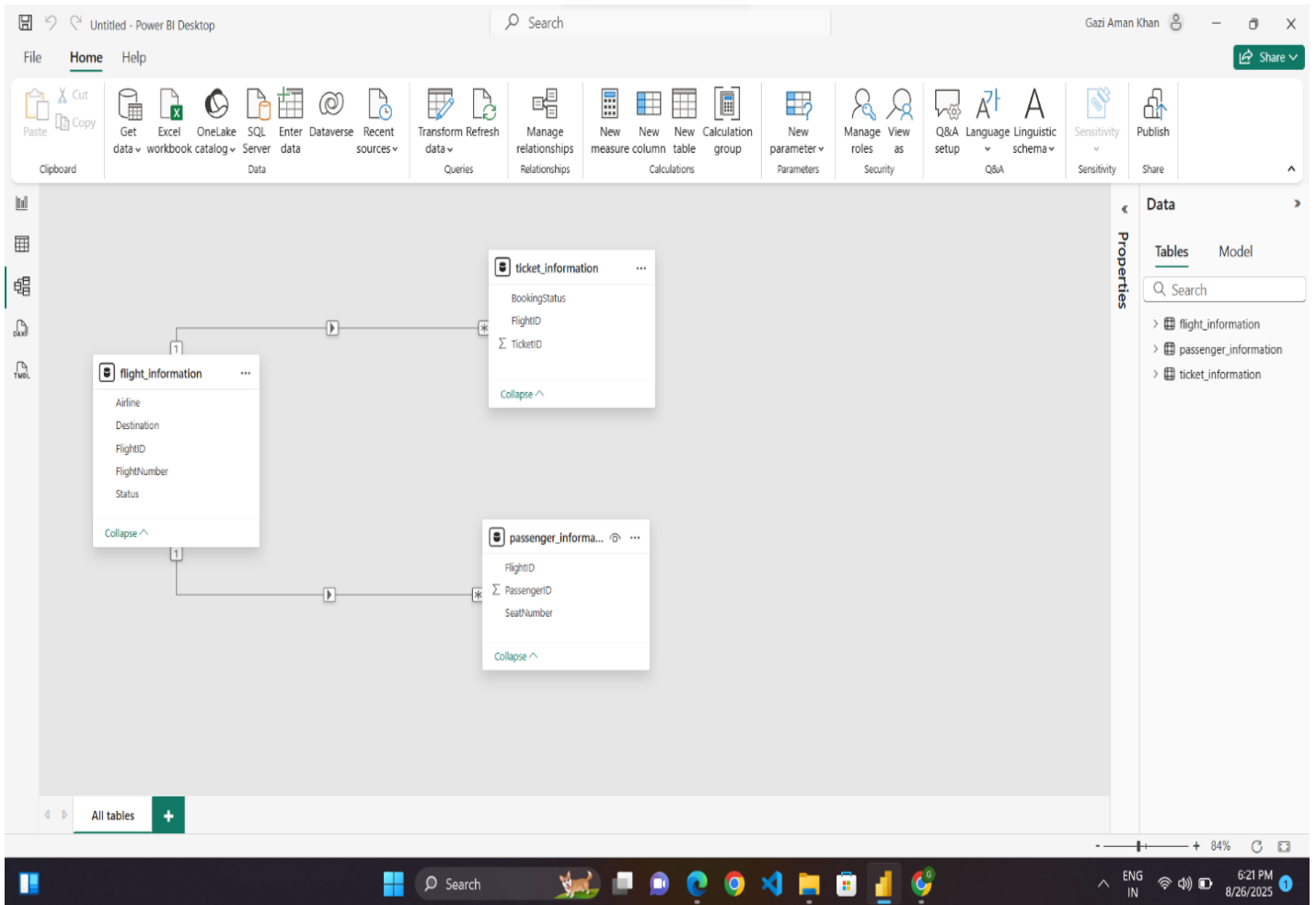
Removed Duplicates

Removed Columns

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Task 2: Data Modeling

- *Created relationships using FlightID.*
- *Ensured correct cardinality (One-to-Many).*



Task 3: Enhanced Data Insights

- Added conditional column: Performance Status = Best / To Be Improved.
- Extracted Flight Numbers using Column From Examples.

Query Settings

Table.AddColumn(#"Filtered Rows1", "Performance status", each if [Status] = "On Time" then "Best" else "To Be Improved")

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name: Performance status

Column Name: Status Operator: equals Value: On Time Output: Best

Else: To Be Improved

OK Cancel

FlightID	FlightNumber	Airline	Destination	Status	Performance status
1019	FL1663	Airline B	Chicago	Cancelled	To Be Improved
1020	FL1130	Airline A	New York	On Time	Best
1021	FL1661	Airline B	New York	Cancelled	To Be Improved
1022	FL1308	Airline A	Houston	Delayed	To Be Improved
1023	FL1769	Airline A	Chicago	On Time	Best
1024	FL1343	Airline B	Chicago	Delayed	To Be Improved
1025	FL1491	Airline D	Phoenix	On Time	Best
1026	FL1413	Airline D	Chicago	Cancelled	To Be Improved

6 COLUMNS, 184 ROWS Column profiling based on top 1000 rows

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Query Settings

Table.TransformColumnTypes(#"Renamed Columns",{{"Extracted flight number", Int64.Type}})

Transform Column Types

Column Name: Extracted flight number Target Type: Int64

OK Cancel

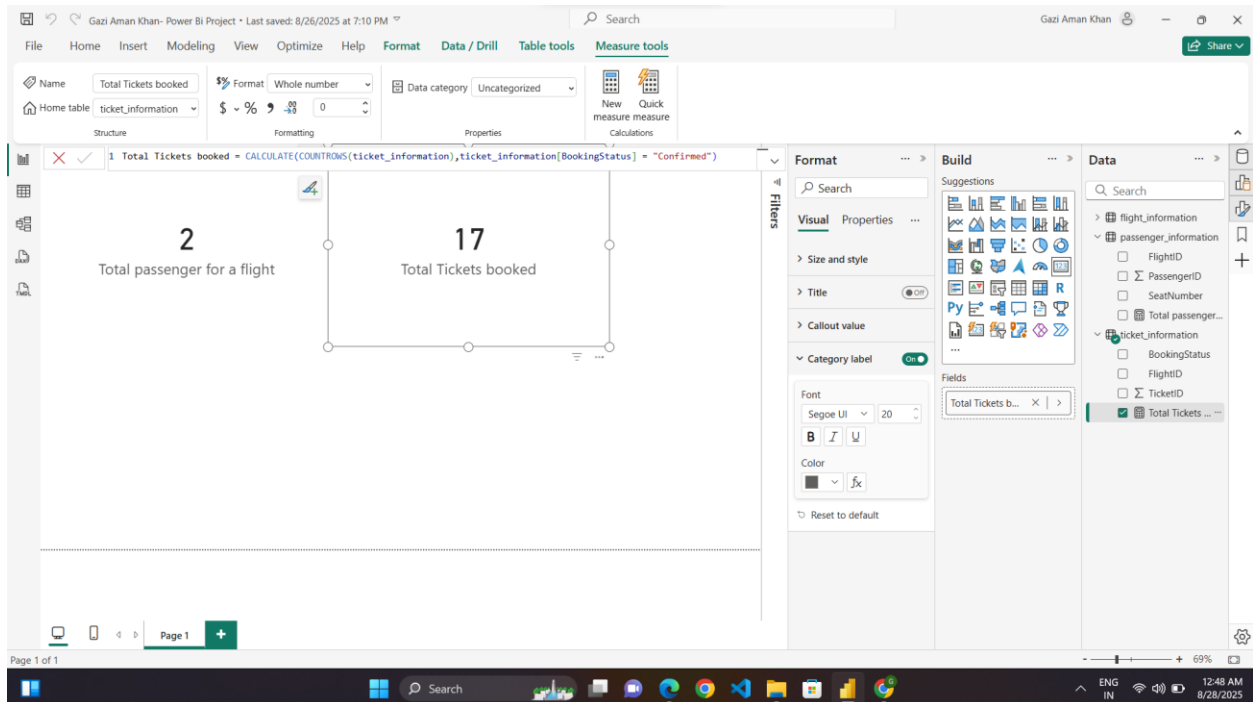
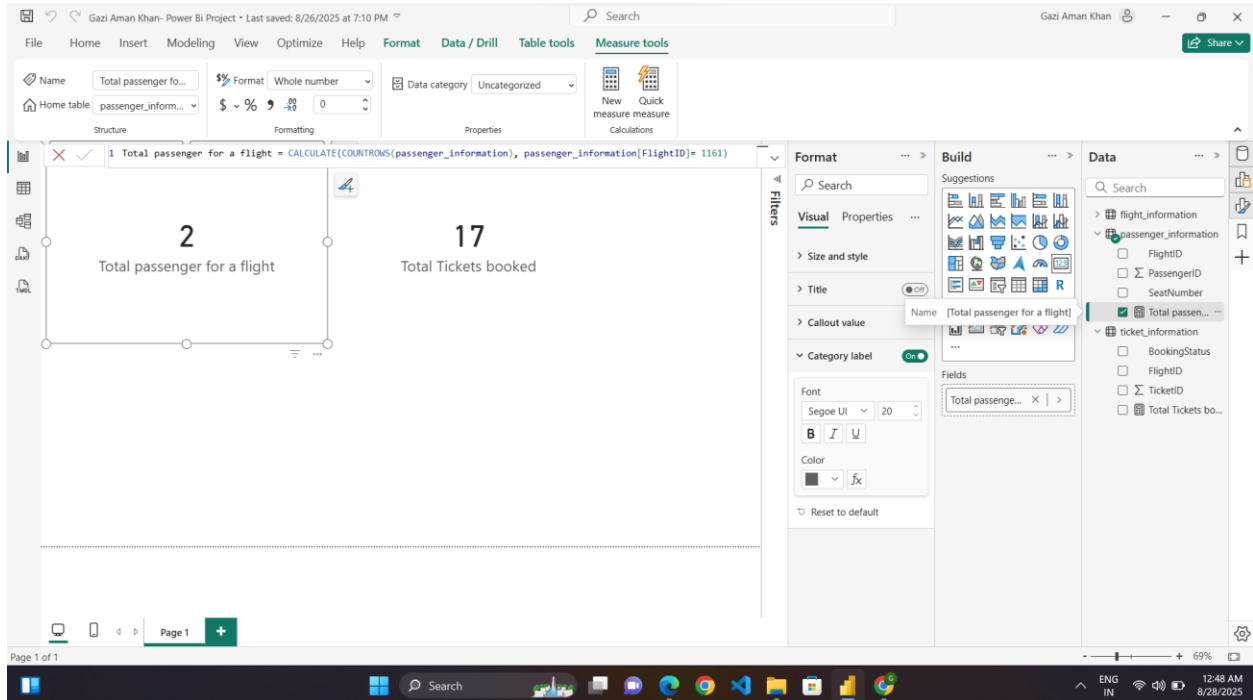
FlightNumber	Airline	Destination	Status	Performance status	Extracted flight number	
1001	FL1102	Airline D	Houston	On Time	Best	1102
1002	FL1435	Airline B	Chicago	On Time	Best	1435
1003	FL1860	Airline A	New York	Cancelled	To Be Improved	1860
1004	FL1270	Airline C	Chicago	Delayed	To Be Improved	1270
1005	FL1106	Airline C	New York	Delayed	To Be Improved	1106
1006	FL1071	Airline A	Phoenix	On Time	Best	1071
1007	FL1700	Airline C	Los Angeles	Cancelled	To Be Improved	1700
1008	FL1020	Airline C	Los Angeles	Delayed	To Be Improved	1020
1009	FL1614	Airline A	Los Angeles	Cancelled	To Be Improved	1614
1010	FL1121	Airline D	Chicago	Cancelled	To Be Improved	1121
1011	FL1466	Airline A	Phoenix	On Time	Best	1466
1012	FL1214	Airline D	New York	Delayed	To Be Improved	1214
1013	FL1330	Airline C	Houston	On Time	Best	1330
1014	FL1458	Airline C	New York	Delayed	To Be Improved	1458
1015	FL1087	Airline C	Houston	Delayed	To Be Improved	1087
1016	FL1372	Airline B	New York	Delayed	To Be Improved	1372
1017	FL1099	Airline D	Phoenix	Delayed	To Be Improved	1099
1018	FL1871	Airline B	Houston	Delayed	To Be Improved	1871
1019	FL1663	Airline B	Chicago	Cancelled	To Be Improved	1663
1020	FL1130	Airline A	New York	On Time	Best	1130
1021	FL1661	Airline B	New York	Cancelled	To Be Improved	1661
1022	FL1308	Airline A	Houston	Delayed	To Be Improved	1308
1023	FL1769	Airline A	Chicago	On Time	Best	1769
1024	FL1343	Airline B	Chicago	Delayed	To Be Improved	1343
1025	FL1491	Airline D	Phoenix	On Time	Best	1491
1026	FL1413	Airline D	Chicago	Cancelled	To Be Improved	1413

7 COLUMNS, 184 ROWS Column profiling based on top 1000 rows

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Task 4: Calculations Using DAX

- *Measure: Total passengers for a specific flight.*
- *Measure: Total tickets booked.*
- *Filtered Table: Best flights only.*



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Search

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Share

FileHomeHelpTable tools

NameBest_Flights

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

StructureRelationshipsCalculationsCalendars

1 Best_Flights = SELECTCOLUMNS(FILTER(flight_information,flight_information[Performance status]= "Best"),"Flight ID",flight_information[FlightID],"Flight Number",flight_information[FlightNumber], "Airline",flight_information[Airline])

Flight IDFlight NumberAirline

1001	FL1102	Airline D
1002	FL1435	Airline B
1006	FL1071	Airline A
1011	FL1466	Airline A
1013	FL1330	Airline C
1020	FL1130	Airline A
1023	FL1769	Airline A
1025	FL1491	Airline D
1027	FL1805	Airline D
1028	FL1385	Airline D
1029	FL1191	Airline D
1030	FL1955	Airline B
1031	FL1276	Airline B
1033	FL1459	Airline D
1034	FL1313	Airline B
1036	FL1252	Airline D
1039	FL1560	Airline B
1043	FL1681	Airline C
1044	FL1475	Airline B
1046	FL1975	Airline D
1048	FL1189	Airline A
1050	FL1686	Airline C
1052	FL1562	Airline D
1053	FL1875	Airline C
1055	FL1243	Airline B
1057	FL1504	Airline A
1060	FL1610	Airline D

Data

Search

Measures (2)

Best_Flights

flight_information

passenger_information

ticket_information

Table: Best_Flights (78 rows)

Search

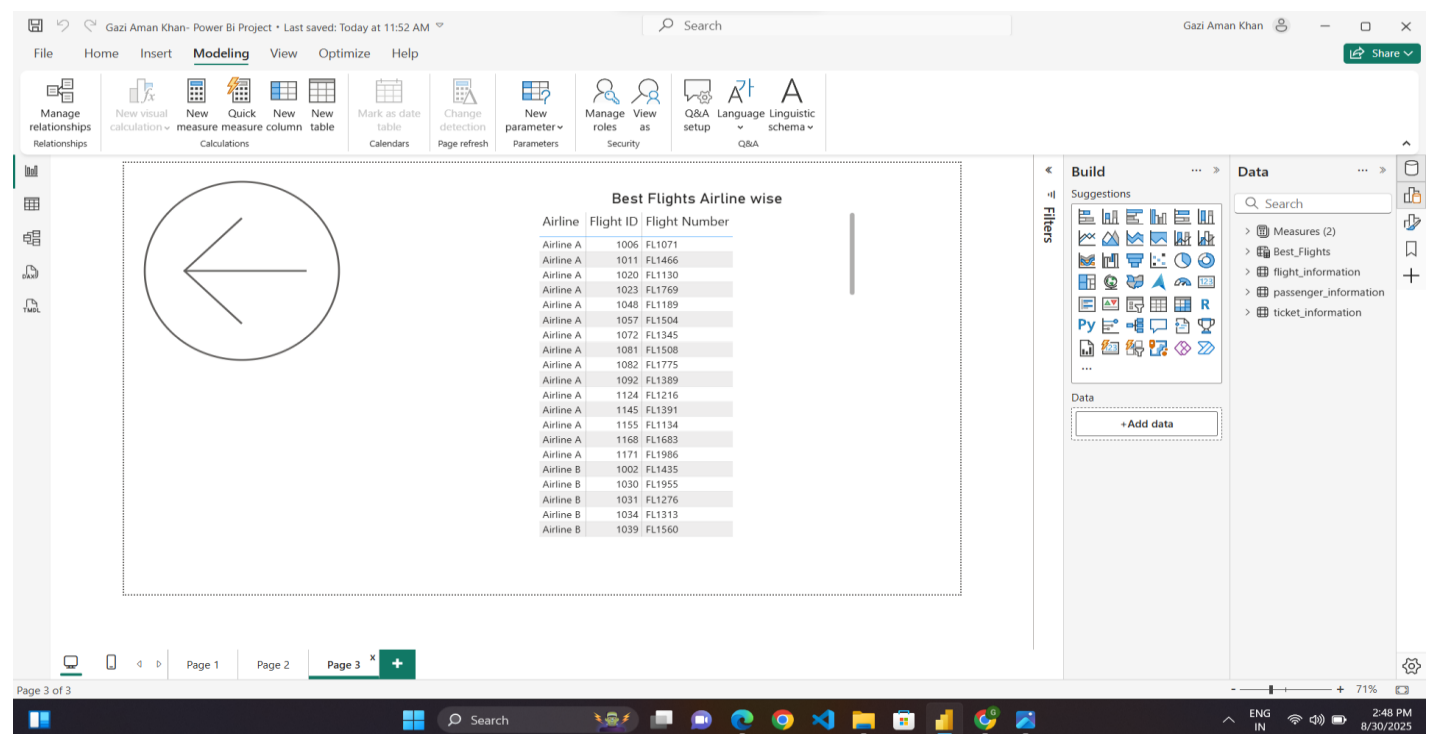
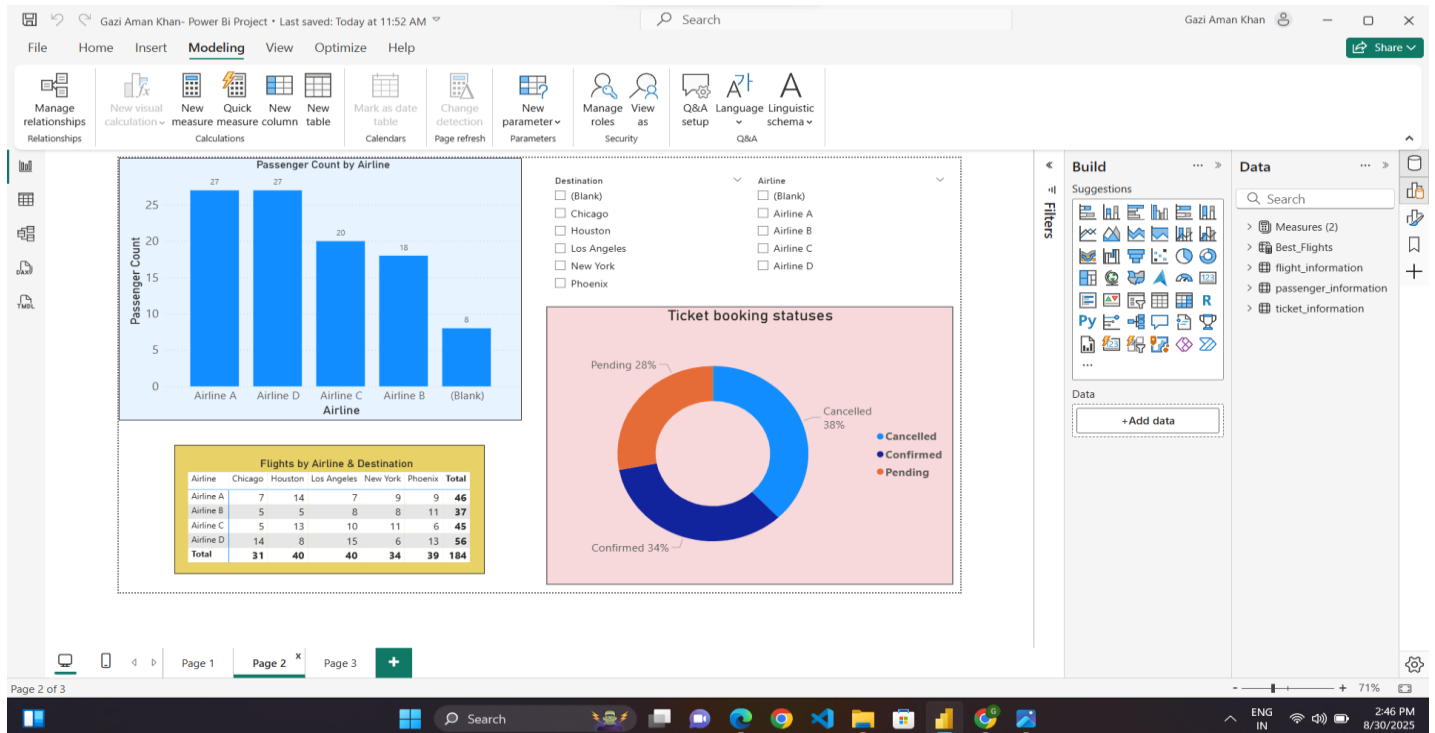
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Flight ID	Flight Number	Airline
1001	FL1102	Airline D
1002	FL1435	Airline B
1006	FL1071	Airline A
1011	FL1466	Airline A
1013	FL1330	Airline C
1020	FL1130	Airline A
1023	FL1769	Airline A
1025	FL1491	Airline D
1027	FL1805	Airline D
1028	FL1385	Airline D
1029	FL1191	Airline D
1030	FL1955	Airline B
1031	FL1276	Airline B
1033	FL1459	Airline D
1034	FL1313	Airline B
1036	FL1252	Airline D
1039	FL1560	Airline B
1043	FL1681	Airline C
1044	FL1475	Airline B
1046	FL1975	Airline D
1048	FL1189	Airline A
1050	FL1686	Airline C
1052	FL1562	Airline D
1053	FL1875	Airline C
1055	FL1243	Airline B
1057	FL1504	Airline A
1060	FL1610	Airline D

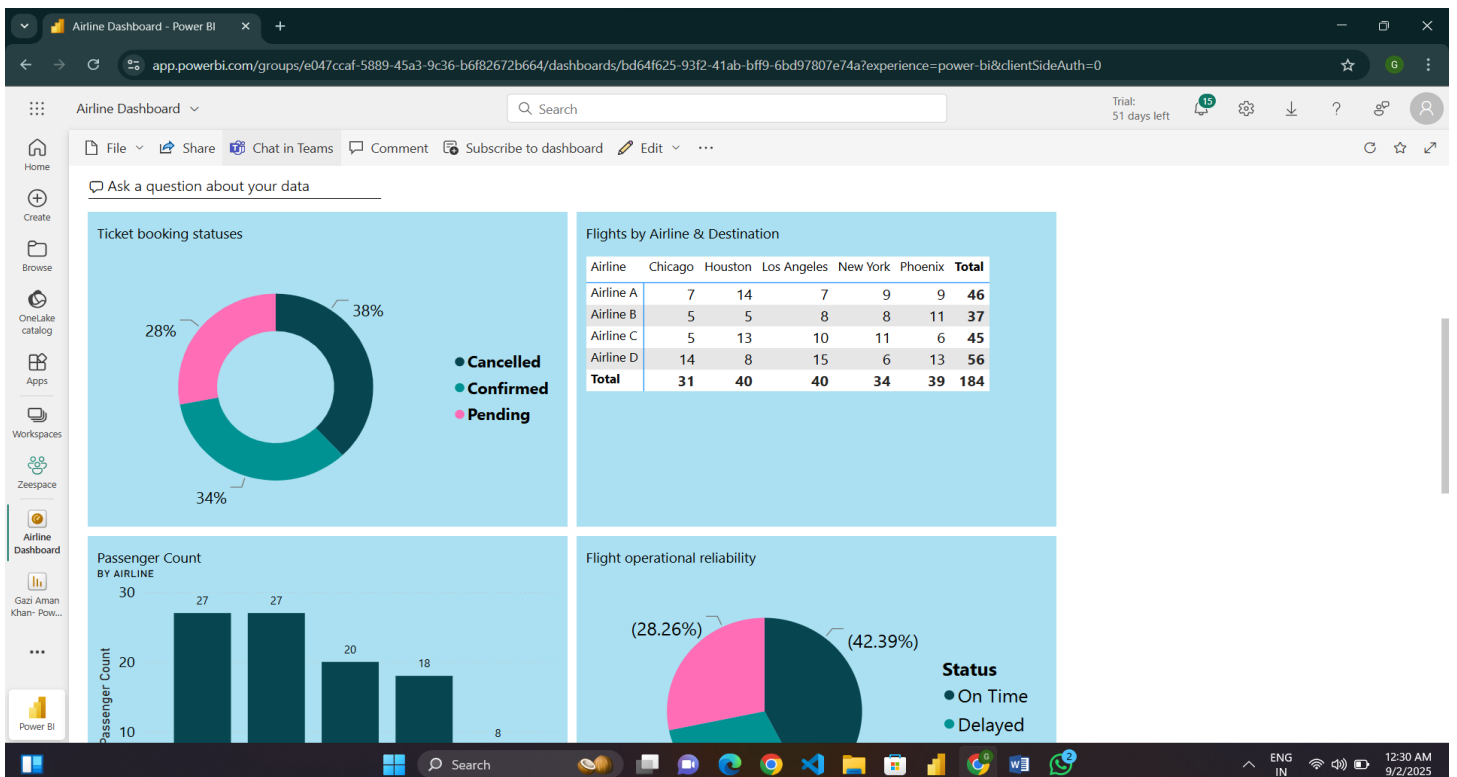
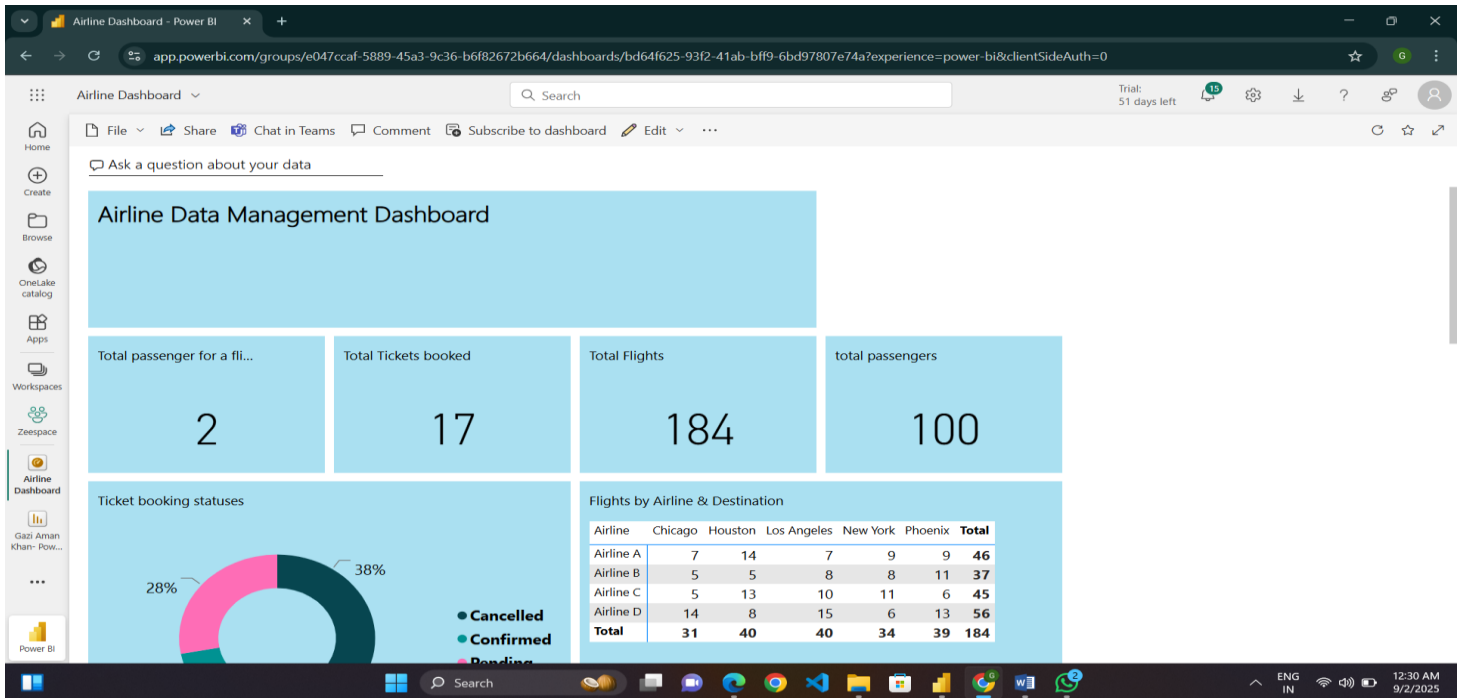
Task 5: Visualizations & Interactivity

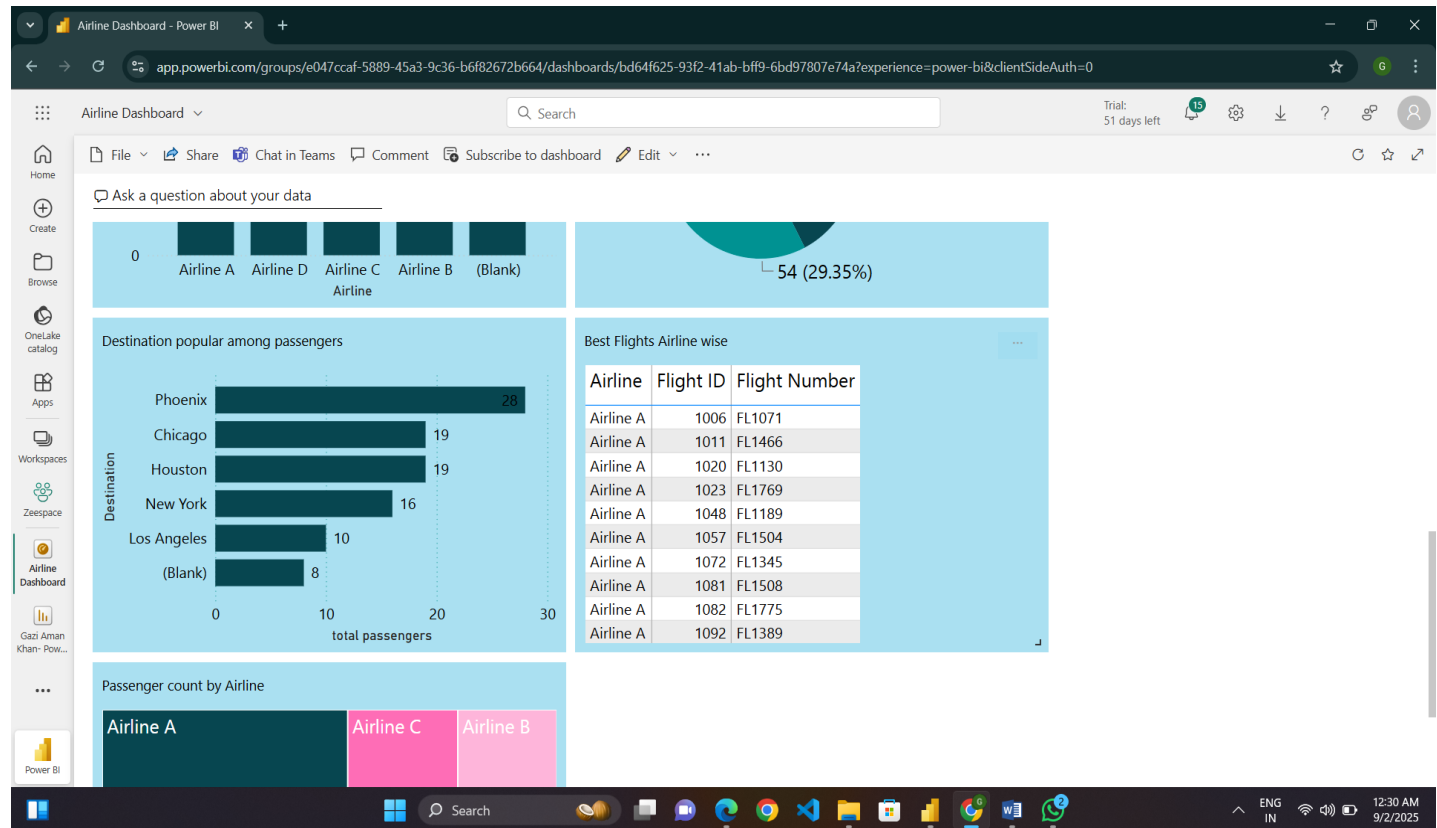
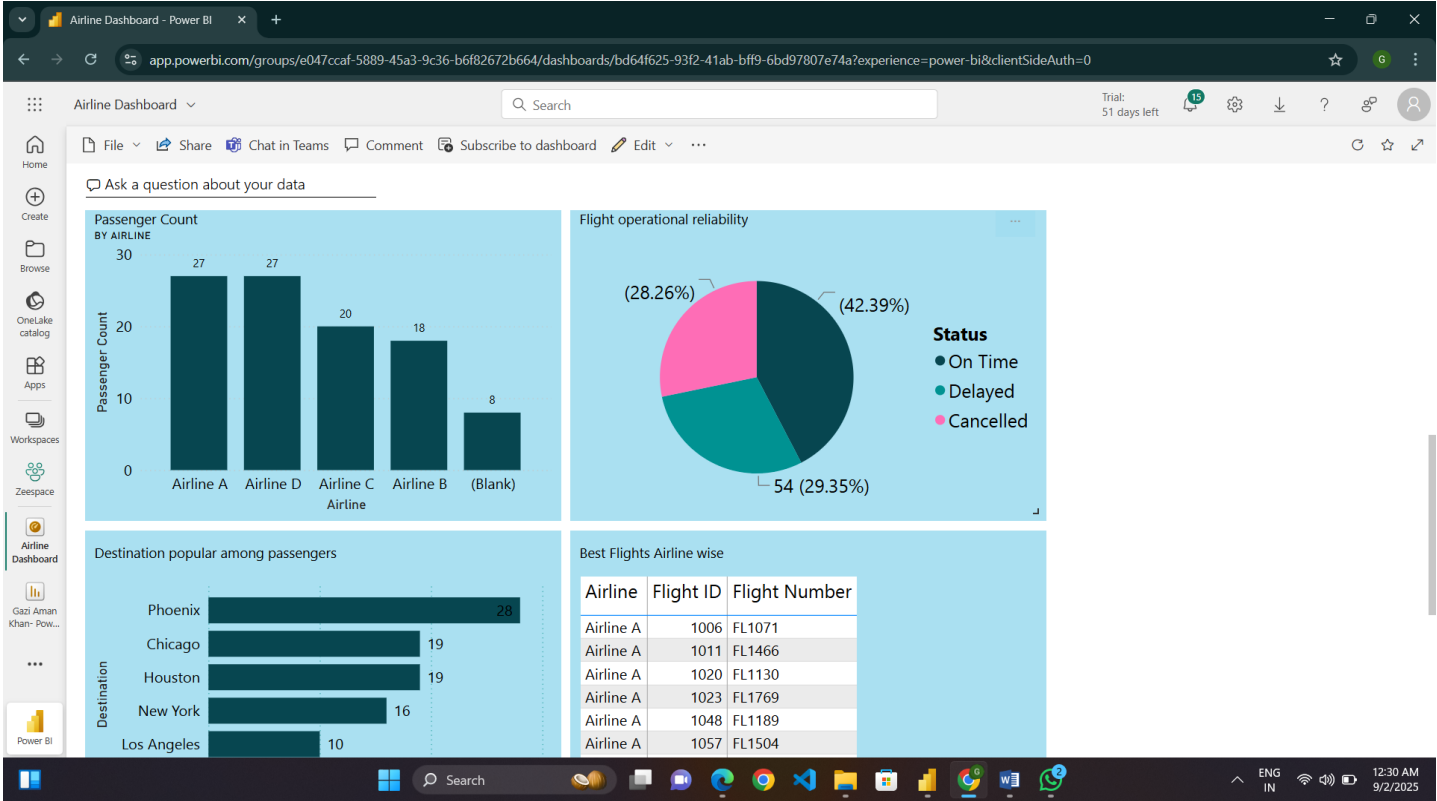
- *Passenger count by Airline (column chart).*
- *Ticket booking statuses (donut chart).*
- *Flights by Airline and Destination (stacked chart/matrix).*
- *Added slicers for Destination & Airline.*
- *Created drillthrough page for airline-specific insight.*



Task 6: Final Dashboard in Power BI Service

- *Designed comprehensive dashboard with KPIs , visuals and detailed tables .*
- *Published to Power BI Service.*
- *Configured Row-Level Security (RLS) for Airline A and assigned user.*





Airline Dashboard - Power BI

app.powerbi.com/groups/e047ccaf-5889-45a3-9c36-b6f82672b664/dashboards/bd64f625-93f2-41ab-bff9-6bd97807e74a?experience=power-bi&clientSideAuth=0

Airline Dashboard

Search

Trial: 51 days left

FileShareChat in TeamsCommentSubscribe to dashboardEdit

Ask a question about your data

Chicago19

Houston19

New York16

Los Angeles10

(Blank)8

total passengers

Airline A	1011	FL1466
Airline A	1020	FL1130
Airline A	1023	FL1769
Airline A	1048	FL1189
A		
A		
A		
A		
Airline A	1082	FL1775
Airline A	1092	FL1389

Passenger count by Airline

Airline A27

Airline D27

Airline C20

Airline B18

(Blank)

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Airline	Airline A	504
Flight ID	1057	345
Flight Number	FL1504	508

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File Home Insert **Modeling** View Optimize Help

Manage relationships Relationships New visual calculation New measure m Calculations

Manage security roles

Create new security roles and use filters to define row-level data restrictions.

Roles

+ New

Airline A roles

Tables

Best_Flights
flight_informat...
Measures (2)
passenger_inf...
ticket_informat...

Rules

Switch to default editor

```
1 flight_information[Airline]="Airline A"
```

Filter the data that this role can see by entering a DAX filter expression that returns a True/False value. For example: [Entity ID] = "Value"

Save Close

Passenger Count

Airline A

Airline	Chicago	Houston	Los Angeles	New York	Phoenix	Total
Airline A	7	14	7	9	9	46
Airline B	5	5	8	8	11	37
Airline C	5	13	10	11	6	45
Airline D	14	8	15	6	13	56
Total	31	40	40	34	39	184

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File Home Insert **Modeling** View Optimize Help

Manage relationships Relationships New visual calculation New measure measure column New table Mark as date table Change detection Page refresh Parameters Manage View as Q&A Language Linguistic schema Q&A

View as roles

☐ None

☐ Other user

☒ Airline A roles

OK Cancel

Passenger Count by Airline

Airline	Passenger Count
Airline A	27
Airline D	27
Airline C	20
Airline B	18
(Blank)	8

Flights by Airline & Destination

Airline	Chicago	Houston	Los Angeles	New York	Phoenix	Total
Airline A	7	14	7	9	9	46
Airline B	5	5	8	8	11	37
Airline C	5	13	10	11	6	45
Airline D	14	8	15	6	13	56
Total	31	40	40	34	39	184

Confirmed 34%

Build

Filters

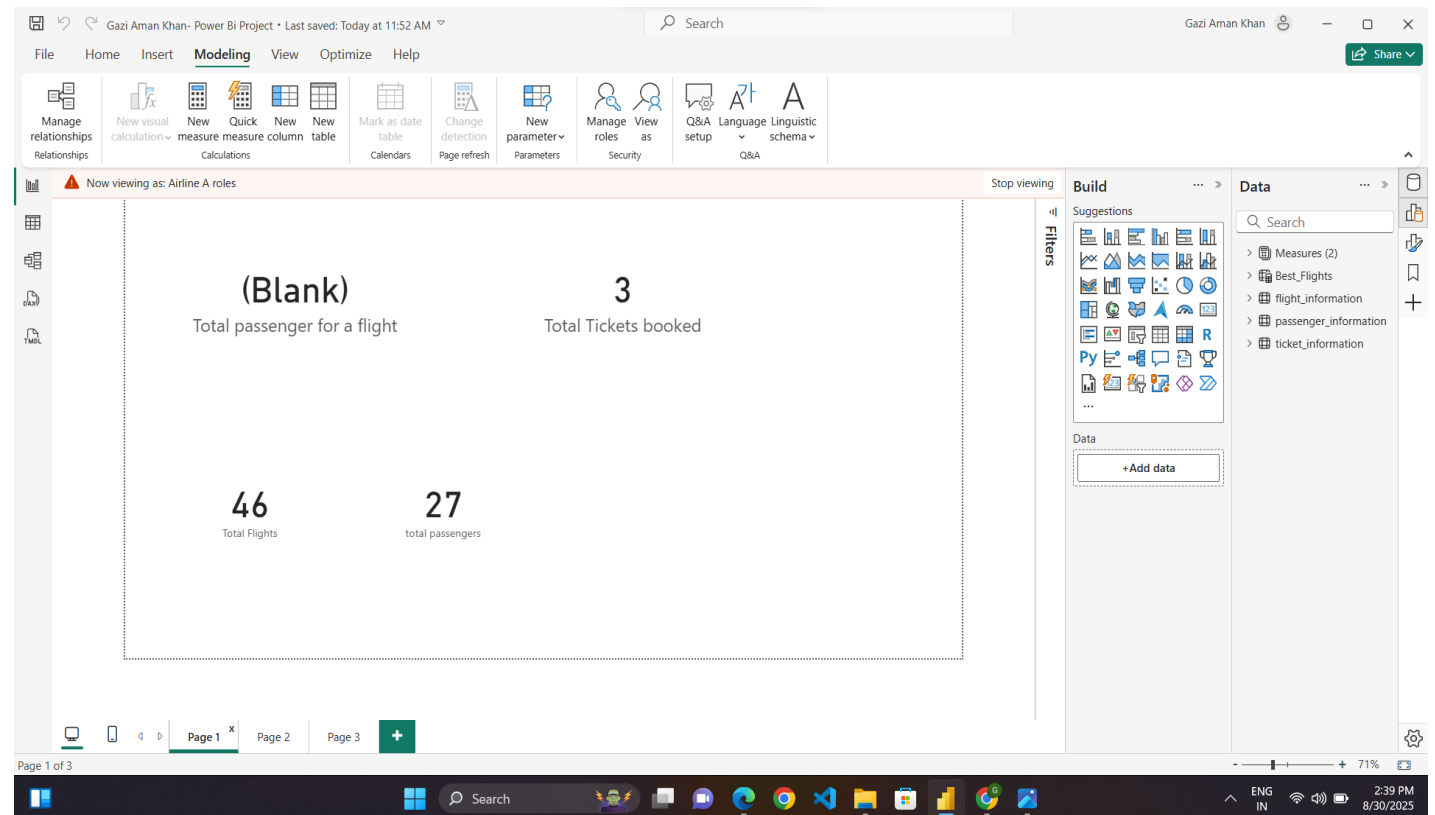
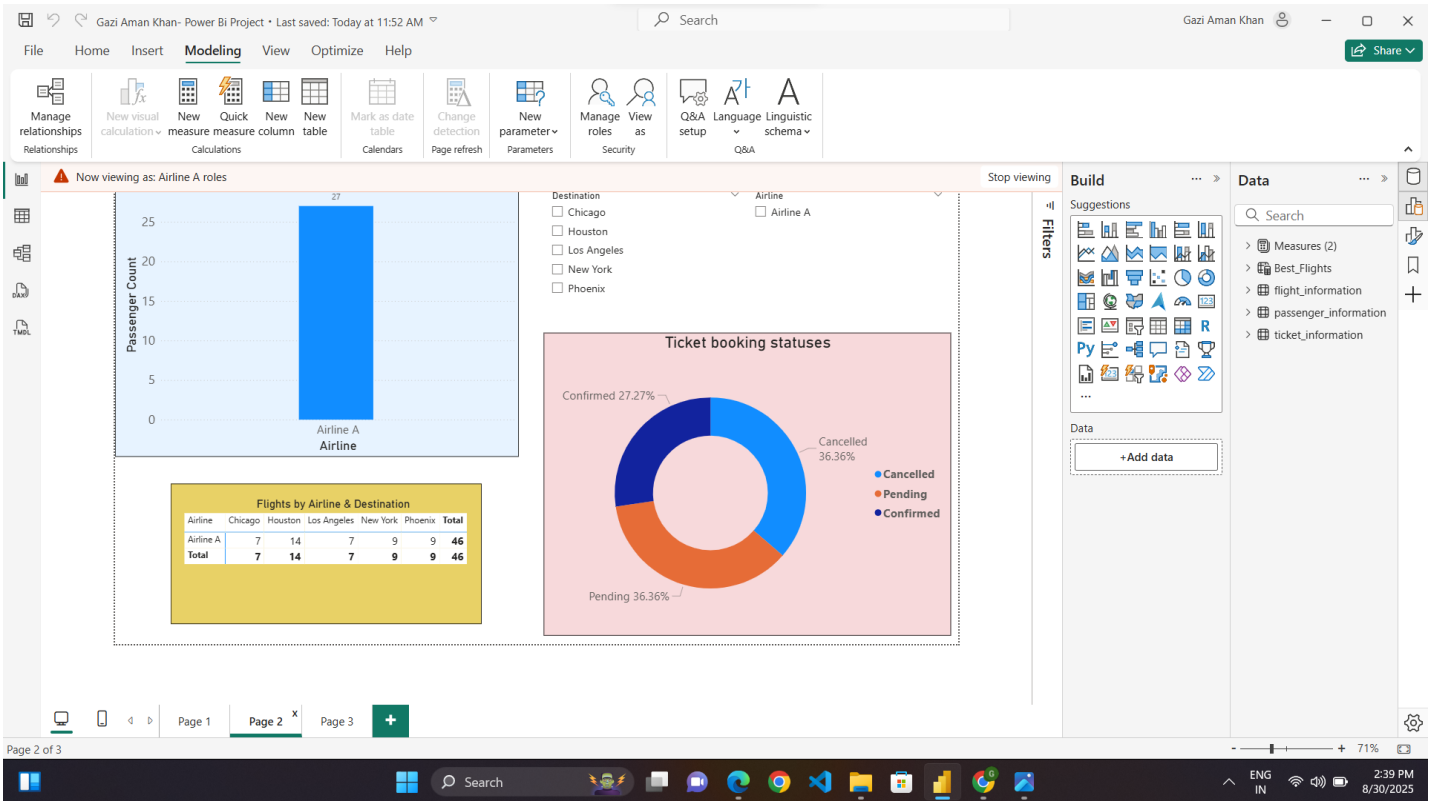
Suggestions

Data

+Add data

Page 2 of 3

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Key Insights & Findings

- Airline B has the highest % of confirmed tickets.
- Airline C has the highest % of canceled tickets.
- Airline D has the highest number of flights.
- Airline B has the lowest number of flights.
- Phoenix is the most popular destination among the passengers.
- Airline A and D has the most number of passengers.
- Airline D has the highest number of best (on time) flights.
- There are total 184 flights and 100 passengers according to the data.
- RLS was successfully applied for **Airline A**, ensuring that users only see data relevant to their airline.
- 42.39% of the flights are on time.

CONCLUSION

This project successfully demonstrated how Power BI can be used for airline data analysis. By cleaning and modeling data, applying DAX calculations, and designing interactive dashboards, I gained key operational insights. Features like RLS ensure secure data sharing, making the solution business-ready.

Project Explanation Video Link:

<https://www.loom.com/share/4d6b9c8566024e9bb07b336b9b860352?sid=623a75e2-aff0-4133-9ba5-eb862b1f80ee>